

Table 3-3. Coverage of Student Outcomes 1-7 by CHE 400-level Technical Elective Courses.

Course	(1) An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.	(2) An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.	(3) An ability to communicate effectively with a range of audiences.	(4) An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.	(5) An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.	(6) An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	(7) An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.
CHE 422	2	3	3	1	1	1	2
CHE 425	4	4	3	3	3	1	4
CHE 426	4	3	4	1	4	3	2
CHE 427	1	3	4	4	4	1	4
CHE 433	4	3	3	3	3	1	1
CHE 451	3	4	4	4	4	1	4
CHE 453	4	1	3	1	1	3	4
CHE 454	4	3	4	1	1	3	4
CHE 455	4	3	3	3	4	4	4
CHE 457	4	3	4	1	1	1	4

4 = topic is main focus of course, grade largely dependent on demonstrated knowledge;

3 = course grade partially dependent on demonstrated knowledge on this topic;

2 = topic introduced in course, course grade not dependent on demonstrated knowledge;

1 = topic not covered specifically in course.

CHE 422. Biochemical Engineering

CHE 425. Nanotechnology for Pharmaceutical Applications

CHE 426. Biopharmaceutical Process Development and Manufacturing

CHE 427. Entrepreneurship in Engineering

CHE 433. Process Simulation With Aspen Plus

CHE 451. Renewable Energy Technologies

CHE 454. Molecular and Macromolecular Engineering

CHE 455. Nanoscale Systems in Chemical Engineering

CHE 457. Colloidal and Interfacial Phenomena